Hyperuniformity of 1D Point Patterns

Week 2 Progress

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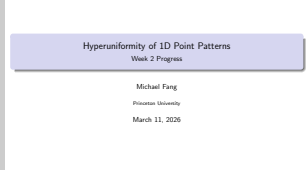

Week 2 progress. We extended the catalog from three metallic-mean chains (all Class I, $\alpha = 3$) to a comprehensive ranking spanning all three hyperuniformity classes, including disordered patterns.

Week 2: Expanding the Catalog

Week 1 recap: Metallic-mean chains (Fibonacci, Silver, Bronze) — all Class I with $\alpha = 3$, measured $\bar{\Lambda}$.

This week: build a comprehensive $(\alpha, \bar{\Lambda})$ ranking spanning all three hyperuniformity classes.

Ordered patterns added:

- Period-doubling chain — Class II
- 0222 chain — Class III

Disordered patterns added:

- Uniformly randomized lattice (URL) Class I
- Stealthy hyperuniform ($\chi = 0.1, 0.2, 0.3$) — Class I

Data sources:

- Substitution chains: $N \approx 10^6$ tiles
- URL: $N = 10^5$, exact $\bar{\Lambda}$ formula
- Stealthy: collab ensemble, ~4,314 configs/ χ, $N = 2{,}000$

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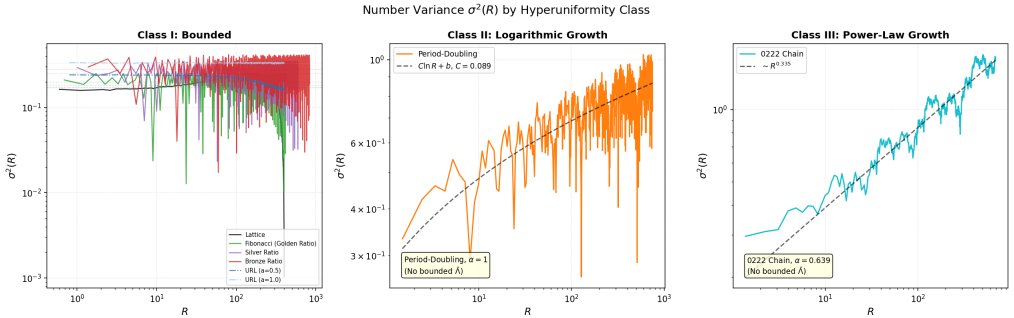
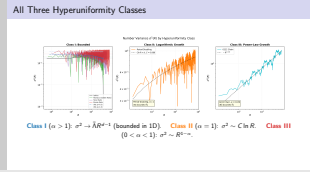
The goal this week was to populate all three hyperuniformity classes and compare disordered vs. ordered patterns at the same α. Collab data: 1D stealthy configurations at $\chi = 0.1, 0.2, 0.3$, each with $N = 2000$ and number density $\rho = 1$.

All Three Hyperuniformity Classes



Class I ($\alpha > 1$): $\sigma^2 \to \bar{\Lambda} R^{d-1}$ (bounded in 1D). **Class II** ($\alpha = 1$): $\sigma^2 \sim C \ln R$. **Class III** ($0 < \alpha < 1$): $\sigma^2 \sim R^{1-\alpha}$.

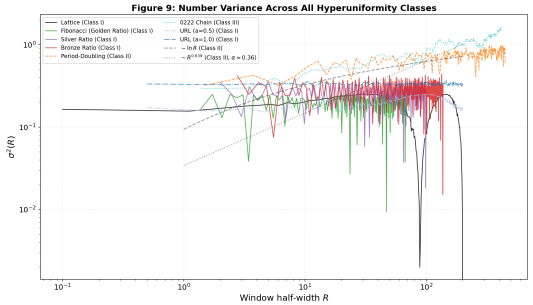


This is the key organizing figure. All Class I patterns asymptote to a constant (bounded variance). Period-doubling is clearly logarithmic. 0222 chain shows power-law growth with exponent between 0 and 1. Poisson (not shown) would grow linearly.

New Substitution Chains: Class II and III

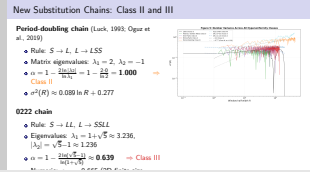
Period-doubling chain (Luck, 1993; Oguz et al., 2019)

- Rule: $S \to L,\ L \to LSS$
- Matrix eigenvalues: $\lambda_1 = 2,\ \lambda_2 = -1$
- $\alpha = 1 - \frac{2\ln|\lambda_2|}{\ln\lambda_1} = 1 - \frac{2\cdot 0}{\ln 2} = \mathbf{1.000}$ $\Rightarrow$ Class II
- $\sigma^2(R) \approx 0.089 \ln R + 0.277$



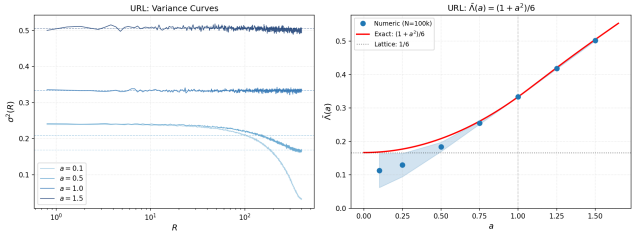
0222 chain

- Rule: $S \to LL,\ L \to SSLL$
- Eigenvalues: $\lambda_1 = 1+\sqrt{5} \approx 3.236$, $|\lambda_2| = \sqrt{5}-1 \approx 1.236$
- $\alpha = 1 - \frac{2\ln(\sqrt{5}-1)}{\ln(1+\sqrt{5})} \approx \mathbf{0.639}$ $\Rightarrow$ Class III



Period-doubling: $|\lambda_2| = 1$ so $\ln|\lambda_2| = 0$ giving exactly $\alpha = 1$. 0222 chain: both eigenvalues exceed 1 in magnitude, giving $0 < \alpha < 1$ (Class III). Note the slight discrepancy $\alpha_{\text{fit}} = 0.665$ vs. theory 0.639 — this is a finite-size effect at $N \approx 10^6$; the power-law regime has not fully saturated.

Uniformly Randomized Lattice (URL)



Each point displaced from lattice site:

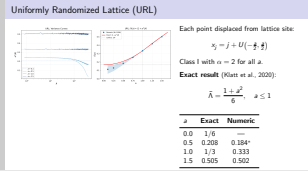
$$x_j = j + U\left(-\frac{a}{2}, \frac{a}{2}\right)$$

Class I with $\alpha = 2$ for all a.

Exact result (Klatt et al., 2020):

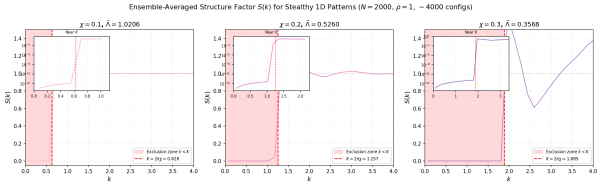
$$\bar{\Lambda} = \frac{1 + a^2}{6}, \quad a \leq 1$$

a	Exact	Numeric
0.0	1/6	—
0.5	0.208	0.184*
1.0	1/3	0.333
1.5	0.505	0.502



The URL is a simple disordered Class I pattern. For $a \leq 1$, no two displaced intervals overlap, and the exact formula $\bar{\Lambda} = (1 + a^2)/6$ holds (Klatt et al. 2020, eq. 12). At $a = 0$: recovers perfect lattice $\bar{\Lambda} = 1/6$. At $a = 1$ ("cloaking" limit): $\bar{\Lambda} = 1/3$, twice the lattice value. For $a < 0.75$, the sliding-window estimator is biased by sampling aliasing (window spacing $\Delta R \approx 0.5$ resolves half the displacement period); we use the exact formula instead.

Stealthy Hyperuniform: Structure Factor



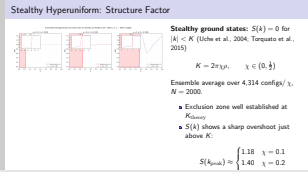
Stealthy ground states: $S(k) = 0$ for $|k| < K$ (Uche et al., 2004; Torquato et al., 2015)

$$K = 2\pi\chi\rho, \qquad \chi \in (0, \tfrac{1}{2})$$

Ensemble average over 4,314 configs/ χ, $N = 2000$.

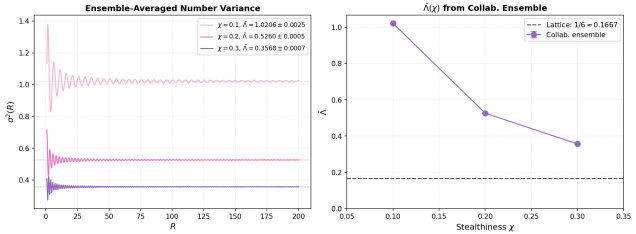
- Exclusion zone well established at K_{theory}
- $S(k)$ shows a sharp overshoot just above K:

$$S(k_{\text{peak}}) \approx \begin{cases} 1.18 & \chi = 0.1 \\ 1.40 & \chi = 0.2 \end{cases}$$



The exclusion zone is exact by construction. The overshoot ("ringing") just above K is a finite-χ effect: the system must compensate for the suppressed low-k modes. This overshoot contributes extra variance and will cause the measured $\bar{\Lambda}$ to systematically exceed the ideal stealthy prediction.

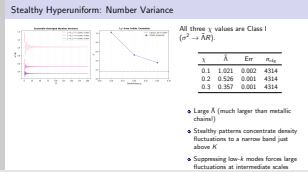
Stealthy Hyperuniform: Number Variance



All three χ values are Class I ($\sigma^2 \to \bar{\Lambda} R$).

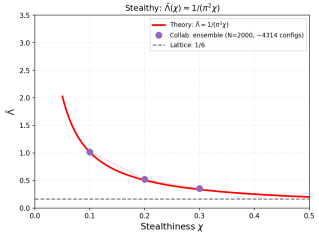
χ	$\bar{\Lambda}$	Err	n_{cfg}
0.1	1.021	0.002	4314
0.2	0.526	0.001	4314
0.3	0.357	0.001	4314

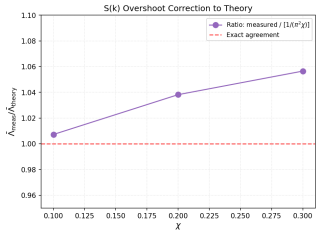
- Large $\bar{\Lambda}$ (much larger than metallic chains!)
- Stealthy patterns concentrate density fluctuations to a narrow band just above K
- Suppressing low-k modes forces large



Bootstrap error: split the last 2/3 of the variance curve into 4 groups, take std of group means. All three are robustly Class I; the variance clearly saturates. The key surprise: stealthy $\bar{\Lambda}$ is much larger than the metallic chains (0.20–0.28), even though both are Class I with bounded variance.

Stealthy Hyperuniform: Analytic Theory





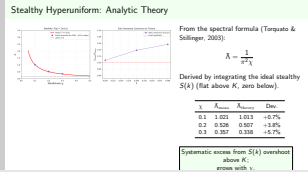
From the spectral formula (Torquato & Stillinger, 2003):

$$\bar{\Lambda} = \frac{1}{\pi^2 \chi}$$

Derived by integrating the ideal stealthy $S(k)$ (flat above K, zero below).

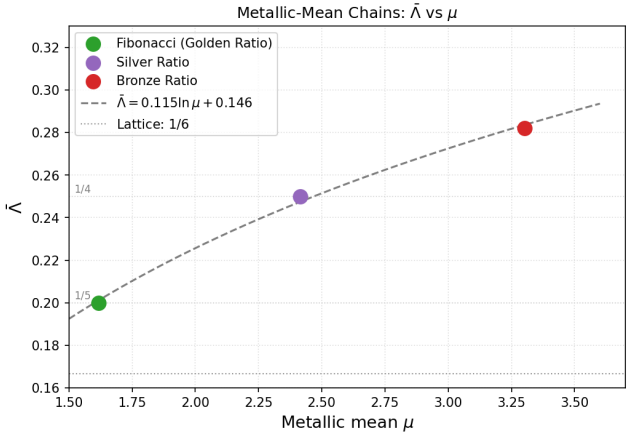
χ	$\bar{\Lambda}_{\text{meas}}$	$\bar{\Lambda}_{\text{theory}}$	Dev.
0.1	1.021	1.013	+0.7%
0.2	0.526	0.507	+3.8%
0.3	0.357	0.338	+5.7%

Systematic excess from $S(k)$ overshoot above K;



Derivation: for ideal stealthy, $S(k) = 0$ for $k < K$ and $S(k) = 1$ for $k > K$. The 1D variance formula gives $\sigma^2(R) \to 2\rho R/(\pi K) = R/(\pi^2\chi)$ at large R, so $\bar{\Lambda} = 1/(\pi^2\chi)$. The small systematic excess (0.7–5.7%) is entirely explained by the S(k) overshoot just above K, which contributes additional variance at intermediate window radii.

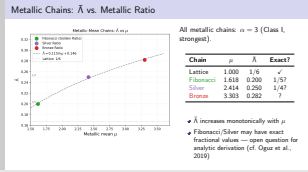
Metallic Chains: $\bar{\Lambda}$ vs. Metallic Ratio



All metallic chains: $\alpha = 3$ (Class I, strongest).

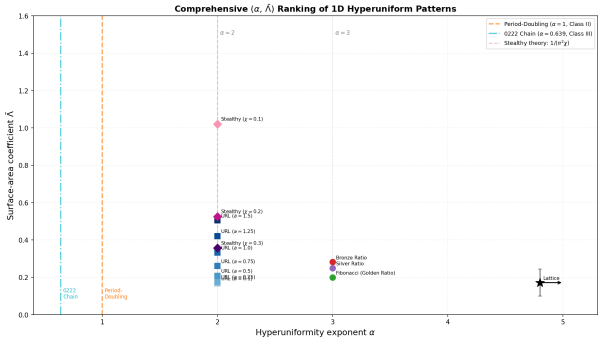
Chain	μ	$\bar{\Lambda}$	Exact?
Lattice	1.000	1/6	✓
Fibonacci	1.618	0.200	1/5?
Silver	2.414	0.250	1/4?
Bronze	3.303	0.282	?

- $\bar{\Lambda}$ increases monotonically with μ
- Fibonacci/Silver may have exact fractional values — open question for analytic derivation (cf. Oguz et al., 2019)



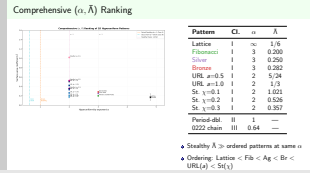
Lattice $\bar{\Lambda} = 1/6$ is exact from Torquato & Stillinger (2003). Fibonacci $\bar{\Lambda} = 0.200$ (our $N \approx 10^6$) matches Zachary & Torquato (2009) to $< 0.1\%$. Silver $\bar{\Lambda} = 0.250$ is within 0.016% of 1/4 — this may be an exact result derivable from the substitution structure, but we have not proven it analytically.

Comprehensive $(\alpha, \bar{\Lambda})$ Ranking



Pattern	Cl.	α	$\bar{\Lambda}$
Lattice	I	∞	1/6
Fibonacci	I	3	0.200
Silver	I	3	0.250
Bronze	I	3	0.282
URL a=0.5	I	2	5/24
URL a=1.0	I	2	1/3
St. χ=0.1	I	2	1.021
St. χ=0.2	I	2	0.526
St. χ=0.3	I	2	0.357
Period-dbl.	II	1	—
0222 chain	III	0.64	—

- Stealthy $\bar{\Lambda} \gg$ ordered patterns at same α
- Ordering: Lattice < Fib < Ag < Br <



Main takeaway: within Class I, both α and $\bar{\Lambda}$ matter. Higher α means faster variance suppression. Lower $\bar{\Lambda}$ means smaller absolute fluctuation amplitude. The lattice is optimal on both counts. Stealthy patterns are Class I ($\alpha = 2$) but have anomalously large $\bar{\Lambda}$ because suppressing $k < K$ modes displaces fluctuations to intermediate scales. URL($a = 0.5$) with $\bar{\Lambda} = 5/24 \approx 0.208$ is comparable to Fibonacci ($\bar{\Lambda} \approx 0.200$) — two very different physical mechanisms giving nearly the same surface-area coefficient.